UCab — Ride with Ease

Full Stack Web Application Project Report (MERN Stack)

1. Project Title

UCab — Cab Booking Web Application

2. Project Description

UCab is a full-stack cab booking web application that allows users to book rides easily using an online platform. The system provides features such as user registration, login, cab browsing, booking rides, and viewing booking history. The application also includes an admin panel where administrators can manage users, bookings, and cab details. This system is developed using the MERN stack which includes MongoDB, Express.js, React.js, and Node.js.

3. Application Scenarios / Use Cases

1. Users can register and log in to the platform to book rides.

2. Users can select available cab types such as Mini, Sedan, SUV, or Luxury.

3. Users can enter pickup location, destination, booking date, and payment method.

4. Admin can manage the fleet of cabs and monitor bookings.

5. Admin dashboard displays total bookings, completed rides, and revenue statistics.

4. Technical Architecture Overview

Component	Role	Technology
Frontend	User interface and interaction	React.js, Bootstrap
Backend	API services and server logic	Node.js, Express.js
Database	Stores user, booking, and cab data	MongoDB
Authentication	Secure login and token management	JWT, bcrypt

File Upload	Uploading cab images	Multer

5. Prerequisites

- Node.js v16 or higher
- MongoDB database
- npm package manager
- Code editor such as VS Code
- Web browser such as Chrome or Firefox

6. Prior Knowledge Required

- Basic knowledge of JavaScript
- Understanding of React.js
- Backend development using Node.js and Express
- Database operations with MongoDB
- REST API concepts

7. Project Objectives

- Develop a web platform for easy cab booking.
- Provide separate modules for users and administrators.
- Allow real-time ride booking and management.
- Implement secure authentication using JWT.
- Manage cab data and bookings through an admin dashboard.

8. System Workflow

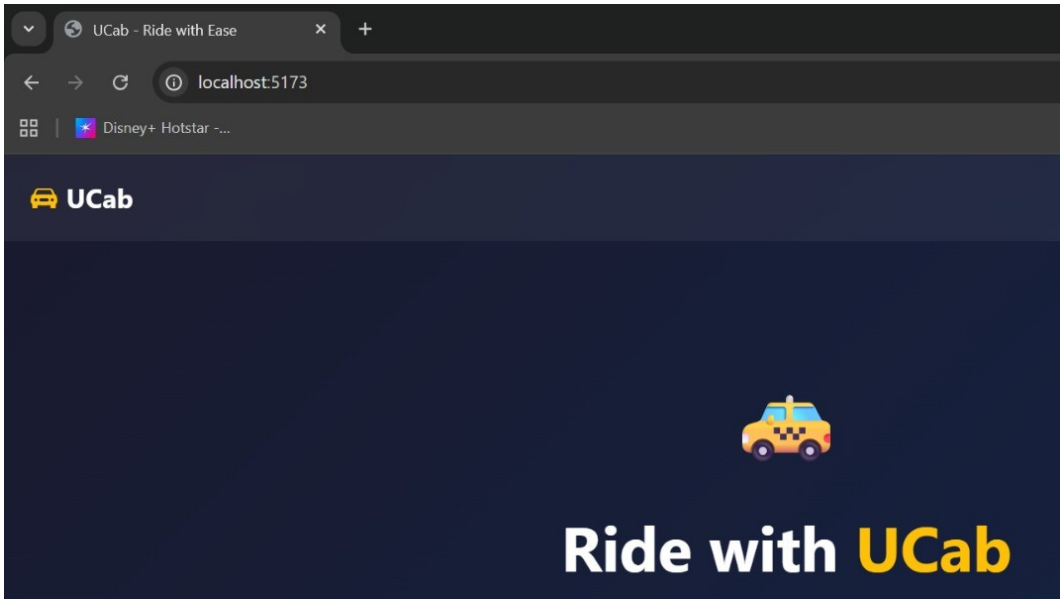
1. User opens the UCab website.
2. User registers or logs in to the system.
3. User selects a cab type and enters ride details.
4. System calculates estimated fare.
5. Booking request is stored in the database.
6. Admin can monitor and manage bookings from the dashboard.

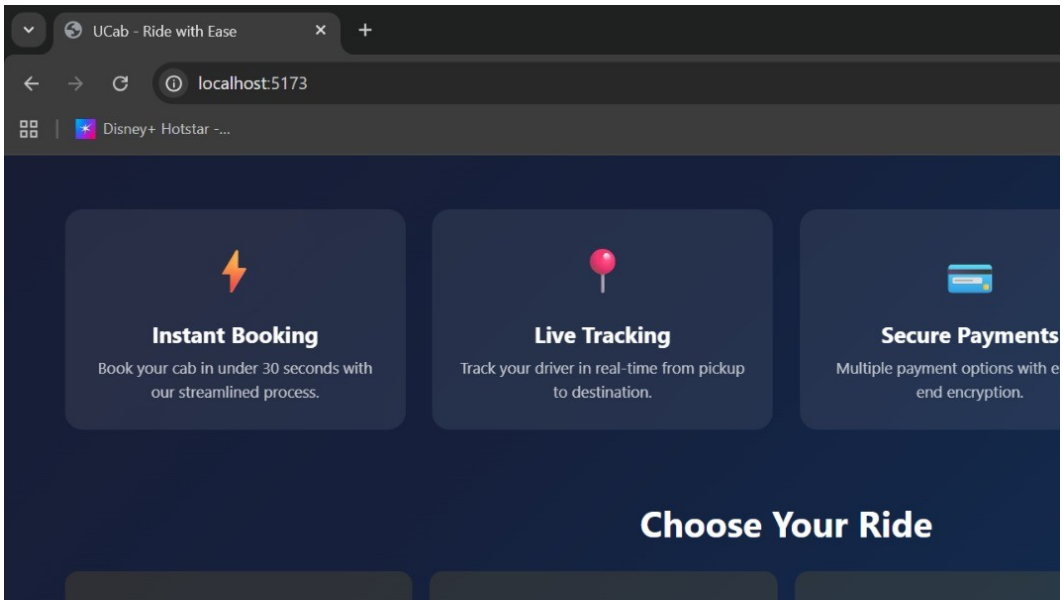
9. Testing & Validation

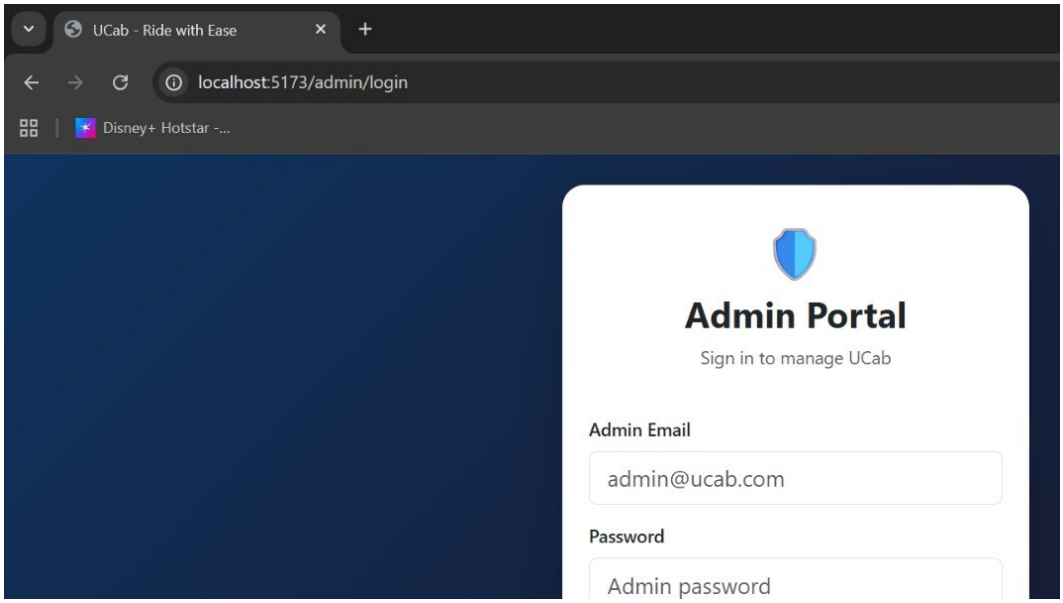
Test Case ID	Input	Expected Output	Status
TC01	User login with valid credentials	Login successful	PASS
TC02	User books a cab	Booking created	PASS
TC03	Admin adds new cab	Cab stored in database	PASS
TC04	User cancels booking	Booking status updated	PASS

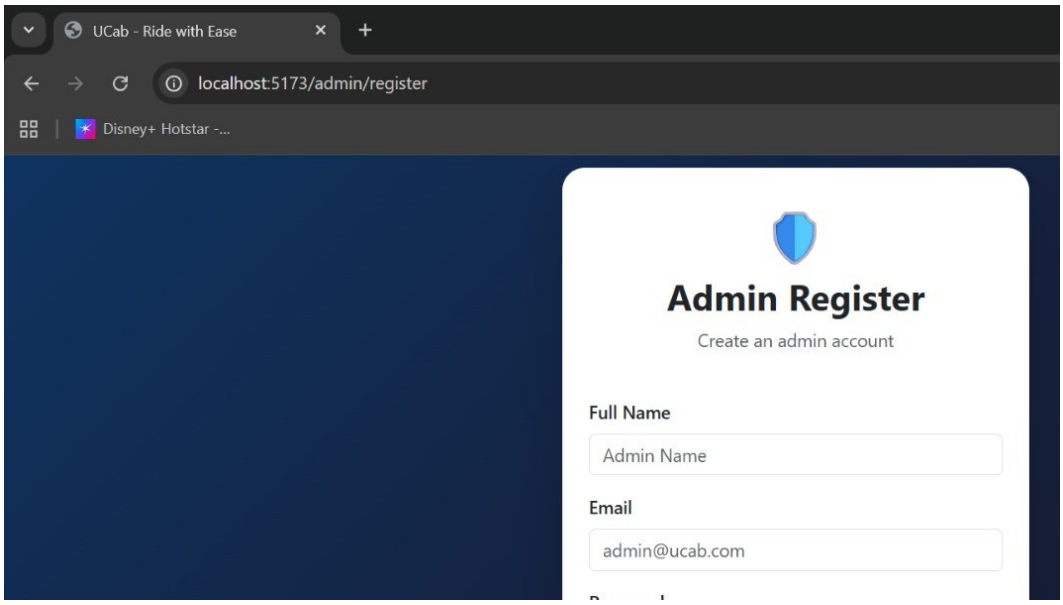
10. Results

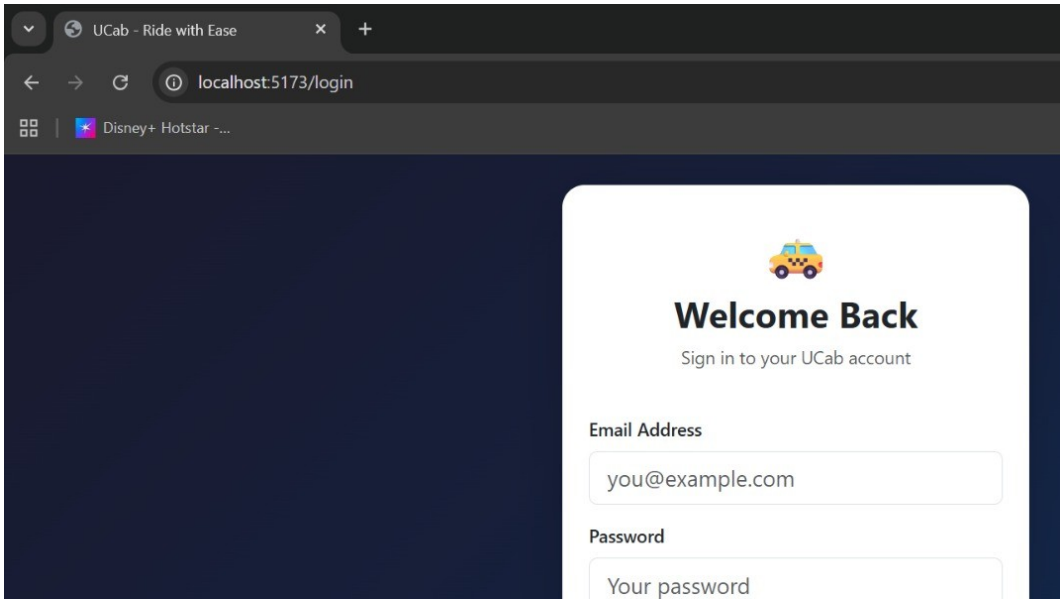
The UCab system successfully allows users to book cabs and administrators to manage operations. The following screenshots show the working system interface.

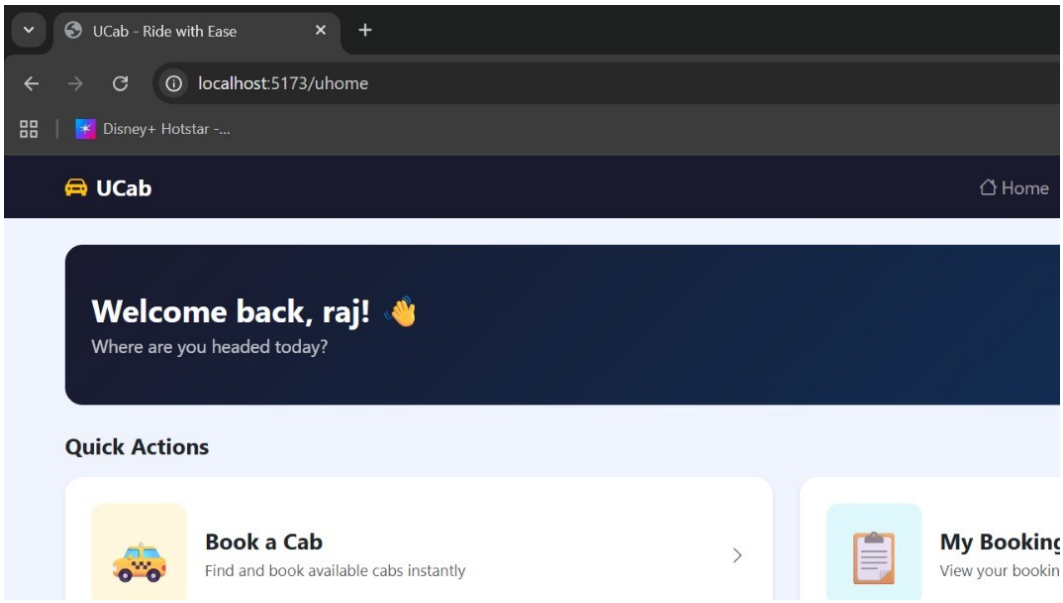


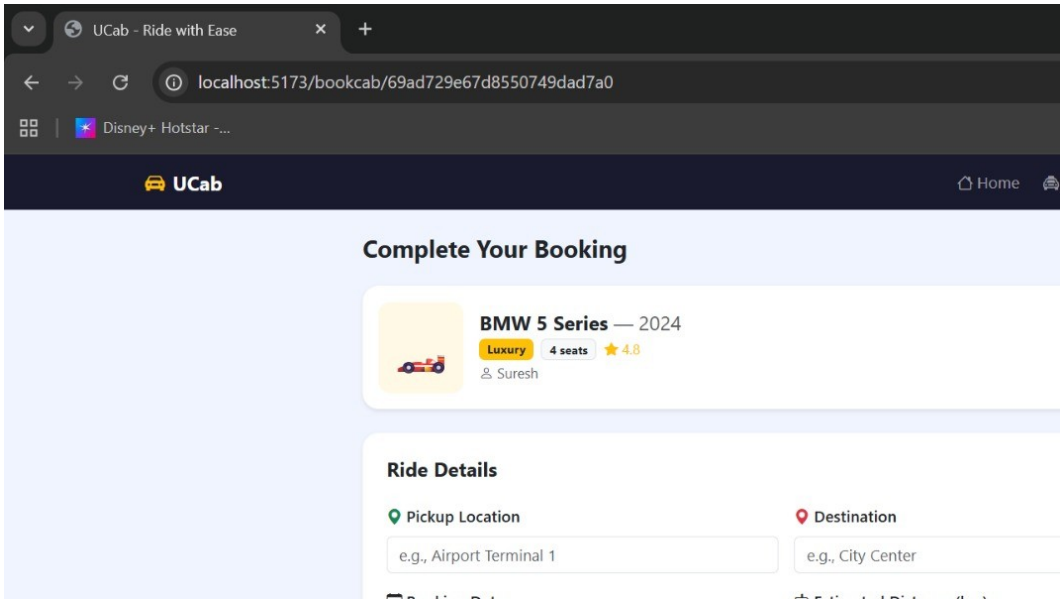

UCab - Ride with Ease
localhost:5173
Disney+ Hotstar -...
Instant Booking
Book your cab in under 30 seconds with our streamlined process.
Live Tracking
Track your driver in real-time from pickup to destination.
Secure Payments
Multiple payment options with e
end encryption.
Choose Your Ride

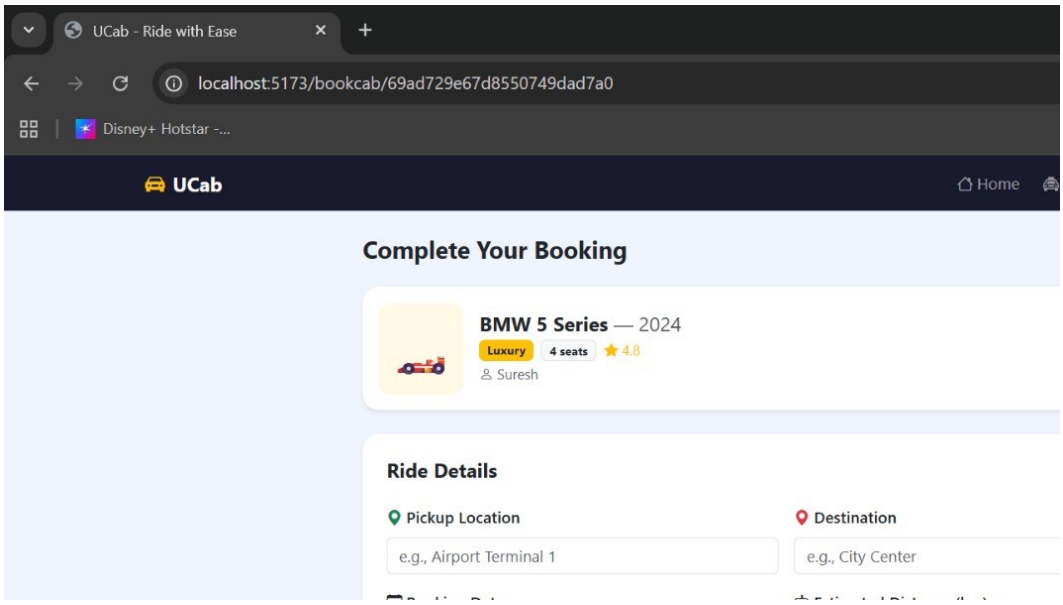

UCab - Ride with Ease
localhost:5173/admin/login
Disney+ Hotstar -...
Admin Portal
Sign in to manage UCab
Admin Email
admin@ucab.com
Password
Admin password

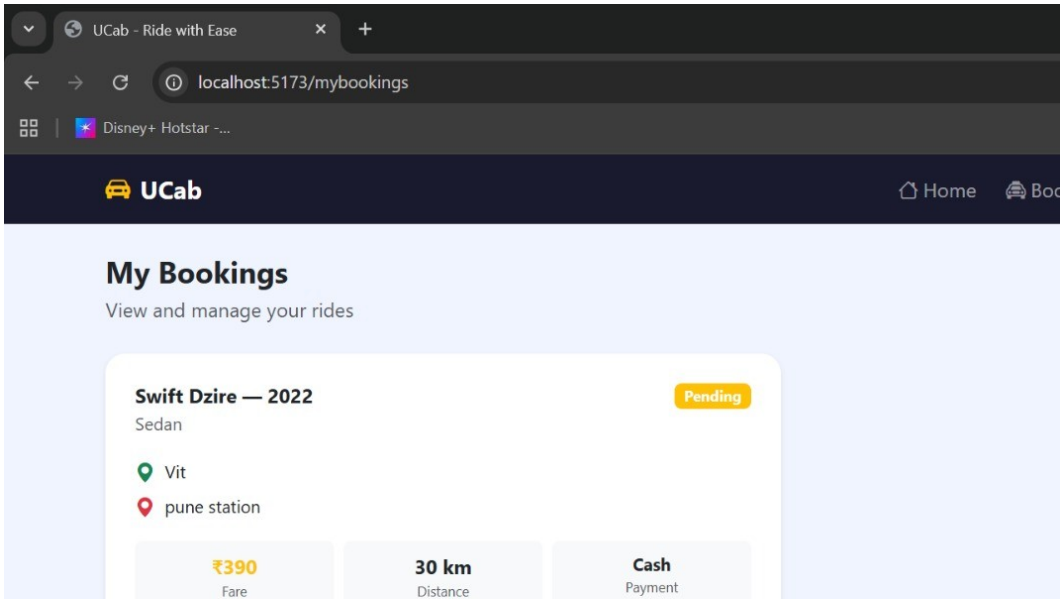

UCab - Ride with Ease
localhost:5173/admin/register
Disney+ Hotstar -...
Admin Register
Create an admin account
Full Name
Admin Name
Email
admin@ucab.com

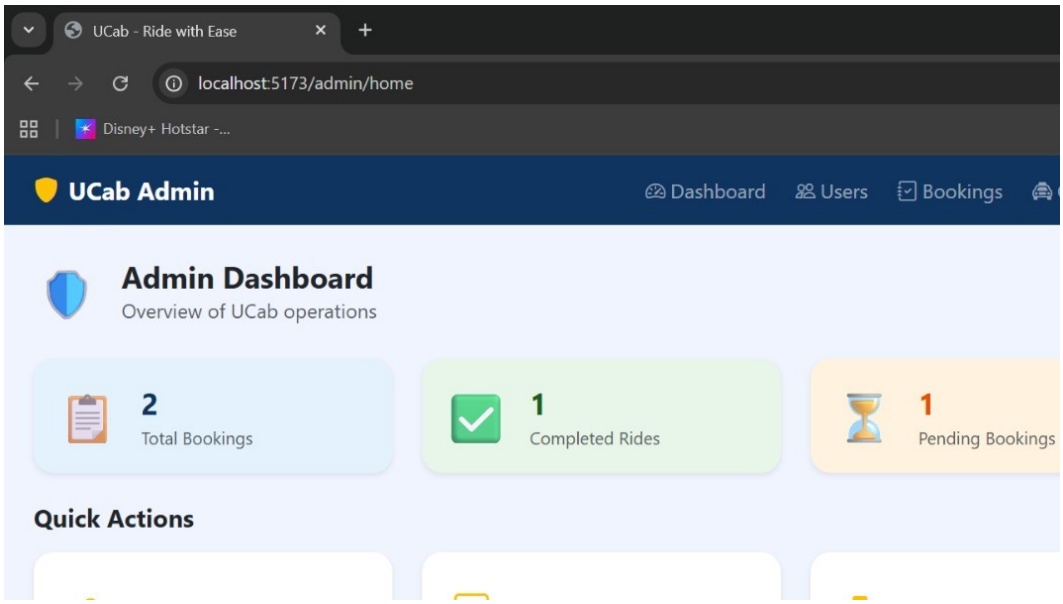

UCab - Ride with Ease
localhost:5173/login
Disney+ Hotstar -...
Welcome Back
Sign in to your UCab account
Email Address
you@example.com
Password
Your password

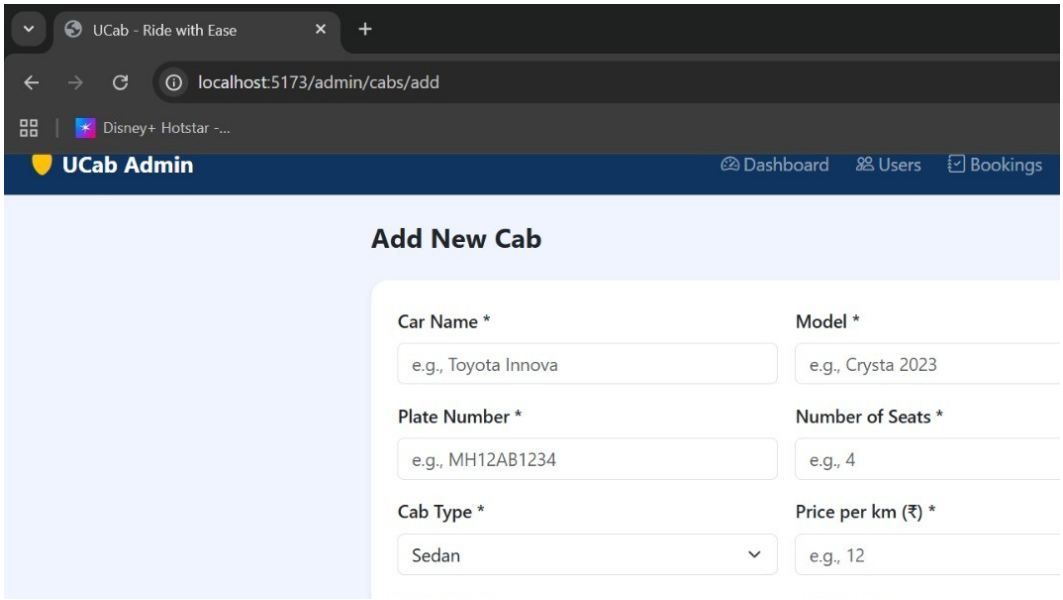

UCab - Ride with Ease
localhost:5173/uhome
Disney+ Hotstar -...
UCab
Home
Welcome back, raj! 👋
Where are you headed today?
Quick Actions
Book a Cab
Find and book available cabs instantly
My Booking
View your bookin


UCab - Ride with Ease
localhost:5173/bookcab/69ad729e67d8550749dad7a0
Disney+ Hotstar -...
UCab
Home
Complete Your Booking
BMW 5 Series — 2024
Luxury
4 seats
4.8
Suresh
Ride Details
Pickup Location
e.g., Airport Terminal 1
Destination
e.g., City Center


UCab - Ride with Ease
localhost:5173/bookcab/69ad729e67d8550749dad7a0
Disney+ Hotstar -...
UCab
Home
Complete Your Booking
BMW 5 Series — 2024
Luxury
4 seats
4.8
Suresh
Ride Details
Pickup Location
e.g., Airport Terminal 1
Destination
e.g., City Center


UCab - Ride with Ease
localhost:5173/mybookings
Disney+ Hotstar -...
UCab
Home
My Bookings
View and manage your rides
Swift Dzire — 2022
Pending
Sedan
Vit
pune station
₹390
Fare
30 km
Distance
Cash
Payment





11. Advantages

• Easy ride booking system.

• User-friendly interface.

• Admin control for fleet management.

• Secure authentication using JWT.

• Real-time booking tracking.

12. Limitations

• Requires internet connection.

• Currently designed for local deployment.

• Payment gateway integration not included.

13. Future Enhancements

• Mobile application for Android and iOS.

• Integration with online payment gateways.

• Real-time GPS tracking of drivers.

• Push notifications for ride updates.

14. Conclusion

The UCab cab booking application demonstrates how modern web technologies can be used to develop a complete ride-booking system. The project successfully integrates frontend, backend, and database technologies to provide a seamless experience for users and administrators.